Mohammad Haider

Software engineer, applied mathematician, and statistician building reliable, inspectable systems and tools.

hi@mhaider.dev | mhaider.dev | linkedin.com/in/haidmoham | github.com/haidmoham

SUMMARY

Software engineer with four years of experience building and validating production data systems, internal tools, and analytics workflows at Microsoft and UVA. I turn loosely defined requirements into reliable, inspectable software; I define what “working” means, measure it, and verify the results before calling a system proven.

PROFESSIONAL EXPERIENCE

University of Virginia | IT Analyst Intermediate, BI Developer **May 2025 - Mar 2026**

- Led analytics projects from stakeholder interviews through deployment and maintenance; coordinated across UVA Facilities Management to turn loosely defined questions and cross-organizational data needs into SQL-backed Tableau dashboards.
- Built and later redesigned the ReUSE Store’s environmental and economic impact dashboard as new workflows emerged; reorganized existing information and added tabs for new decisions. The dashboard recorded at least 400 views.

Microsoft | Software Engineer I **Jul 2021 - Jul 2024**

- Owned the end-to-end migration of a financial reporting cube from SSAS Multidimensional to an established Tabular architecture; kept the legacy service available through four weeks of pipeline monitoring and parity checks before cutover.
- Wrote and deployed Azure Data Factory verification scripts; monitored daily pipeline runs and parity checks to confirm that data moved correctly and remained consistent between the legacy and Tabular services.
- Delivered the cube migration and validated downstream financial-reporting workflows that showed up to 200x faster filtering and approximately 4x faster Excel loading.
- Built a full-stack C#/.NET prototype that replaced a research-heavy manual configuration process with a guided workflow; explained choices in the UI, generated configuration files, and validated the output.

Microsoft | Software Engineering Intern | Summers 2017, 2018

- Implemented a C# service that determined each OneDrive user’s most recently used items for a product UI feature; validated it in development.

SELECTED ENGINEERING WORK

C-1N + Robotics Test Bench | Python, MuJoCo, Simulation | mhaider.dev/c1n

- Used Robotics Test Bench experiments to diagnose load-induced sag in C-1N’s original two-joint legs; added a third coxa degree of freedom that allowed the legs to brace outward.
- Made “standing” measurable with a reproducible static-support baseline; verified a deterministic 10-second rollout with six contacts, at least 0.2346 m of COM-projection margin, at least 0.4495 m of torso height, and numerically zero torso angular speed.
- Built a 33-case external-disturbance suite with 200 ms horizontal force pulses across eight bearings; recorded contact, support margin, and torso response, and kept all eight failed-recovery cases at a force equal to the robot’s weight (1 mg) as the explicit boundary.

TECHNICAL SKILLS

Languages: Python, C#

Data, tools, and platforms: SQL, Kusto (KQL), Tableau, Azure Data Factory, SSAS Multidimensional/Tabular, Git

Simulation and scientific computing: NumPy, pandas, Matplotlib, MuJoCo, Jupyter

EDUCATION

Virginia Tech | B.S. in Statistics **2021**

Coursework: Bayesian Statistics, Experimental Design, Statistical Computing, Mathematical Modeling, Combinatorics, and Proofs